Intraclass Correlation Coefficient (ICC)

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Introduction

The Intraclass Correlation Coefficient (ICC) quantifies inter-rater reliability on continuous measurements. Unlike Pearson's correlation, it penalises systematic rater bias (two raters who differ by a constant still have Pearson = 1 but ICC < 1). Several ICC forms reflect different study designs and questions.

Prerequisites

Variance components; random vs fixed effects.

Theory

Shrout-Fleiss (1979) forms: - **ICC(1, 1)**: one-way random effects; each subject rated by a different random rater. - **ICC(2, 1)**: two-way random effects; **absolute agreement** between raters. - **ICC(3, 1)**: two-way mixed effects; **consistency** (ignores systematic rater bias).

Single rater vs average of k raters: ICC(x, 1) vs ICC(x, k).

Assumptions

Subjects and raters are drawn from appropriate populations; ratings are independent given subject; ICC form matches the intended use.

R Implementation

```r
library(psych)

set.seed(2026)
n <- 30
subject_re <- rnorm(n, 0, 1)

# 3 raters, each with own systematic bias
r1 <- subject_re + rnorm(n, 0, 0.4)
r2 <- subject_re + 0.3 + rnorm(n, 0, 0.4)    # rater 2 higher by 0.3
r3 <- subject_re - 0.2 + rnorm(n, 0, 0.4)

df <- cbind(r1, r2, r3)

icc_res <- ICC(df, lmer = FALSE)
icc_res$results[, c("type", "ICC", "lower bound", "upper bound")]
```

Output & Results

Six ICC forms with 95 % CIs. Agreement ICC (2, 1) is lower than consistency ICC (3, 1) when raters have systematic biases.

Interpretation

"Single-rater absolute-agreement ICC(2, 1) = 0.78 (95 % CI 0.63-0.88); consistency ICC(3, 1) = 0.83. Systematic rater differences moderately lowered absolute agreement."

Practical Tips

- Choose ICC form by study question:
  - **ICC(2, 1)** if you want to quantify agreement including systematic rater differences.
  - **ICC(3, 1)** if you will remove systematic rater effects in practice (e.g., calibrate each rater).
- For clinical usability, average-of-k-raters ICCs are the most interpretable.
- Thresholds (Koo-Li): < 0.5 poor, 0.5-0.75 moderate, 0.75-0.9 good, > 0.9 excellent.
- ICC < 0.7 is usually insufficient for individual decision-making.
- Pair ICC with Bland-Altman for graphical assessment of rater-specific bias.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale